

DATA SCIENCE INTERNSHIP • CAPSTONE PROJECT

AIRLINE FLIGHT DELAY & OPERATIONAL PERFORMANCE ANALYTICS

Analysis of 5.82 million US flights • 2015 Flight Delays & Cancellations

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PROJECT TITLE	Airline Flight Delay & Operational Performance Analytics
TOOLS USED	Python • Pandas • NumPy • Matplotlib • Seaborn • Power BI • Kaggle
DATASET	2015 US Flight Delays and Cancellations Dataset
SUBMISSION DATE	September 2026

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1. Introduction

Air transportation is an important part of modern transportation and business. Flight delays and cancellations can affect passengers, airlines, airports, crew scheduling, aircraft utilization, and overall operational efficiency.

The purpose of this project is to analyze flight operations and identify patterns in flight delays, cancellations, airlines, airports, routes, months, and delay causes.

The project uses a large US flight dataset containing approximately **5.82 million flight records** from 2015.

Python was used for data loading, cleaning, analysis, and exploratory data analysis. Power BI was then used to transform the analytical results into an interactive dashboard for easier visualization and interpretation.

2. Problem Statement

Flight delays can occur because of several factors, including:

- Late arrival of previous aircraft
- Airline-related problems
- Air traffic/system delays
- Weather conditions
- Security-related issues

Analyzing these factors can help identify patterns in operational performance.

The main problem addressed by this project is:

How can historical flight data be analyzed to identify patterns in delays, cancellations, airline performance, airport performance, routes, and major causes of delays?

3. Project Objectives

The main objectives of this project are:

1. Analyze a large US flight dataset containing approximately 5.82 million records.
2. Identify the overall percentage of delayed and cancelled flights.
3. Calculate average departure and arrival delays.
4. Compare delay rates across different airlines.
5. Identify the major causes of flight delays.
6. Analyze monthly patterns in delays and cancellations.
7. Identify the busiest airports.
8. Identify airports with high delay rates.
9. Analyze the busiest flight routes.

10. Build an interactive Power BI dashboard to present the findings.

4. Dataset Description

The dataset used in this project is the **2015 US Flight Delays and Cancellations** dataset obtained through Kaggle.

The dataset contains:

- **5,819,079** flight records
- **31** columns

The major fields include:

Column	Description
YEAR	Year of flight
MONTH	Month of flight
DAY	Day of flight
DAY_OF_WEEK	Day of week
AIRLINE	Airline code
FLIGHT_NUMBER	Flight number
TAIL_NUMBER	Aircraft tail number
ORIGIN_AIRPORT	Origin airport
DESTINATION_AIRPORT	Destination airport
SCHEDULED_DEPARTURE	Scheduled departure time
DEPARTURE_TIME	Actual departure time
DEPARTURE_DELAY	Departure delay in minutes
TAXI_OUT	Taxi-out time
WHEELS_OFF	Wheels-off time
SCHEDULED_TIME	Scheduled flight duration
ELAPSED_TIME	Actual elapsed time
AIR_TIME	Time spent in air
DISTANCE	Flight distance
WHEELS_ON	Wheels-on time
TAXI_IN	Taxi-in time
SCHEDULED_ARRIVAL	Scheduled arrival
ARRIVAL_TIME	Actual arrival
ARRIVAL_DELAY	Arrival delay in minutes
DIVERTED	Whether the flight was diverted

Column	Description
CANCELLED	Whether the flight was cancelled
CANCELLATION_REASON	Reason for cancellation
AIR_SYSTEM_DELAY	Air system delay
SECURITY_DELAY	Security delay
AIRLINE_DELAY	Airline-related delay
LATE_AIRCRAFT_DELAY	Delay caused by late aircraft
WEATHER_DELAY	Weather-related delay

5. Tools and Technologies

Python

Python was used for data processing and exploratory analysis. The main libraries used were:

- **Pandas** — data manipulation and analysis
- **NumPy** — numerical calculations
- **Matplotlib** — data visualization
- **Seaborn** — statistical visualization

Kaggle

Kaggle Notebook was used as the environment for loading and analyzing the large dataset.

Power BI

Power BI was used to create the final interactive dashboard and present the analytical results visually.

GitHub

GitHub was used to store and present the project files, Python notebook, dataset-related files, README, and dashboard screenshots.

6. Data Loading and Exploration

The dataset was loaded using Pandas. During loading, Python generated a *DtypeWarning* for two columns because they contained mixed data types. The dataset nevertheless loaded successfully.

Dataset dimensions

Dimension	Value
Rows	5,819,079
Columns	31

The columns were examined to understand the structure of the dataset before beginning the analysis. The dataset contains information about flight schedules, actual flight operations, delays, cancellations,

airports, airlines, and delay causes.

7. Data Cleaning and Preparation

Data quality was examined by calculating missing values for each column. Several columns contained missing values. The largest number of missing values occurred in:

Column	Missing Values	Missing %
CANCELLATION_REASON	5,729,195	98.46%
LATE_AIRCRAFT_DELAY	4,755,640	81.72%
WEATHER_DELAY	4,755,640	81.72%
AIRLINE_DELAY	4,755,640	81.72%
AIR_SYSTEM_DELAY	4,755,640	81.72%
SECURITY_DELAY	4,755,640	81.72%
ELAPSED_TIME	105,071	1.81%
AIR_TIME	105,071	1.81%
ARRIVAL_DELAY	105,071	1.81%
DEPARTURE_DELAY	86,153	1.48%

The high percentage of missing values in cancellation and delay-cause columns is expected because these fields are not applicable to every flight. For example, a flight that was not delayed would generally not have values recorded for individual delay causes.

The data was therefore analyzed according to the availability and meaning of each field rather than treating every missing value as an error.

8. Exploratory Data Analysis

Several analytical measures were created to understand operational performance. The analysis focused on:

- Flight status
- Delay rates
- Average delays
- Airline performance
- Delay causes
- Monthly trends
- Airport activity
- Airport delay rates
- Flight routes

These analyses were subsequently represented through charts and visualizations in Power BI.

9. Flight Status Analysis

Flights were categorized into different operational statuses based on the available flight information. The corrected results were:

Flight Status	Number of Flights	Percentage
On Time	3,627,112	62.33%
Delayed	2,086,896	35.86%
Cancelled	89,884	1.54%
Unknown	15,187	0.26%

The results show that the majority of flights in the dataset were classified as on time, while a substantial portion experienced delays.

10. Average Delay Analysis

The overall average delays were calculated.

Metric	Average
Average Departure Delay	9.37 minutes
Average Arrival Delay	4.41 minutes

The difference between departure and arrival delay indicates that some flights were able to recover part of their delay during the journey.

11. Airline Performance Analysis

The project compared flight delay performance across airlines.

Average Arrival Delay by Airline

Airline	Average Arrival Delay
NK	14.47 min
F9	12.50 min
B6	6.68 min
EV	6.59 min
MQ	6.46 min
OO	5.85 min
UA	5.43 min
VX	4.74 min
WN	4.37 min
US	3.71 min
AA	3.45 min
HA	2.02 min
DL	0.19 min
AS	-0.98 min

A negative average arrival delay indicates that, on average, flights arrived slightly earlier than their scheduled arrival times.

Delay Rate by Airline

Airline	Total Flights	Delayed Flights	Delay Rate
NK	115,193	56,887	49.38%
F9	90,090	41,232	45.77%
HA	76,041	30,179	39.69%
VX	61,248	24,180	39.48%
US	194,223	76,285	39.28%
B6	262,042	101,998	38.92%
OO	576,814	222,435	38.56%
EV	554,752	213,217	38.43%
WN	1,242,403	470,767	37.89%
MQ	278,791	103,505	37.13%
UA	507,762	186,227	36.68%
AA	712,935	252,191	35.37%
AS	171,439	56,953	33.22%
DL	870,275	250,840	28.82%

This comparison demonstrates differences in delay rates and average arrival delays across airlines.

12. Delay Cause Analysis

The project analyzed the total number of delay minutes associated with each recorded delay cause.

Total Delay Minutes by Cause

Delay Cause	Total Delay Minutes
Late Aircraft	24,961,931
Airline	20,172,956
Air System	14,335,762
Weather	3,100,233
Security	80,985

Late aircraft delay represented the largest total number of recorded delay minutes, followed by airline-related and air-system delays.

Number of Flights Affected

Delay Cause	Affected Flights
Airline Delay	570,022
Air System Delay	564,826
Late Aircraft Delay	556,953
Weather Delay	64,716
Security Delay	3,484

This analysis helps distinguish between the amount of delay time and the number of flights affected.

13. Monthly Delay and Cancellation Analysis

Monthly flight performance was analyzed using total flights, delayed flights, cancelled flights, delay rate, and cancellation rate.

Monthly Delay Rate

Month	Delay Rate
January	38.96%
February	40.88%
March	37.70%
April	35.42%
May	35.25%
June	41.08%
July	38.35%
August	35.43%
September	28.70%
October	29.05%
November	31.93%
December	37.45%

The monthly analysis shows that delay rates varied throughout the year. The highest recorded monthly delay rate was approximately **41.08% in June**, while **September** had the lowest at approximately **28.70%**.

Cancellation Rate

Cancellation rates also varied by month. February had a relatively high cancellation rate of approximately **4.78%**, while September had a substantially lower cancellation rate of approximately **0.45%**.

14. Airport Analysis

14.1 Busiest Origin Airports

The top 15 origin airports by number of flights were identified.

Rank	Airport	Total Flights
1	ATL	346,836
2	ORD	285,884
3	DFW	239,551
4	DEN	196,055
5	LAX	194,673
6	SFO	148,008
7	PHX	146,815
8	IAH	146,622
9	LAS	133,181
10	MSP	112,117
11	MCO	110,982
12	SEA	110,899
13	DTW	108,500
14	BOS	107,847
15	EWR	101,772

ATL recorded the highest number of flights among the analyzed origin airports.

14.2 Airports by Delay Rate

The project also analyzed airports based on their delay rates.

Airport	Total Flights	Delayed Flights	Delay Rate
DAL	58,672	26,612	45.36%
OAK	41,669	18,409	44.18%
HOU	51,132	21,963	42.95%
CLT	99,052	41,441	41.84%
DEN	193,402	80,756	41.76%
IAH	144,019	59,824	41.54%
LAX	192,003	79,661	41.49%
BWI	84,329	34,979	41.48%
ORD	276,554	113,281	40.96%
MIA	68,301	27,720	40.59%
MDW	78,685	31,879	40.51%
LAS	131,937	52,897	40.09%
PHX	145,552	58,272	40.04%
DFW	232,647	93,054	40.00%
PBI	22,134	8,843	39.95%

This analysis provides a different perspective from airport traffic volume because an airport can have a high number of flights without necessarily having the highest delay rate.

15. Flight Route Analysis

The project analyzed flight routes using origin and destination airports.

Top 15 Busiest Flight Routes

Origin	Destination	Total Flights	Delayed Flights	Delay Rate
SFO	LAX	13,400	6,431	47.99%
LAX	SFO	13,109	5,845	44.59%
JFK	LAX	11,853	3,857	32.54%
LAX	JFK	11,851	3,972	33.52%
LAS	LAX	9,651	4,974	51.54%
LAX	LAS	9,522	4,195	44.06%
LGA	ORD	9,179	3,154	34.36%
ORD	LGA	9,101	3,640	40.00%
SFO	JFK	8,306	2,701	32.52%
JFK	SFO	8,301	2,798	33.71%
OGG	HNL	8,268	3,765	45.54%
HNL	OGG	8,246	3,277	39.74%
ATL	MCO	8,148	3,129	38.40%
MCO	ATL	8,144	2,436	29.91%
LAX	ORD	8,131	2,995	36.83%

The results show that several high-volume routes also experienced substantial numbers of delayed flights.

16. Power BI Dashboard

The analysis was converted into an interactive Power BI dashboard. The dashboard provides a visual summary of the major findings from the Python analysis.

Dashboard components include:

- Total flight statistics
- Flight status distribution
- Airline delay analysis
- Monthly delay trends
- Monthly cancellation trends
- Delay causes
- Busiest airports
- Airport delay rates
- Busiest flight routes
- Route delay rates

The Power BI dashboard makes it easier to explore the results and identify operational patterns.

Dashboard Screenshot — Overview

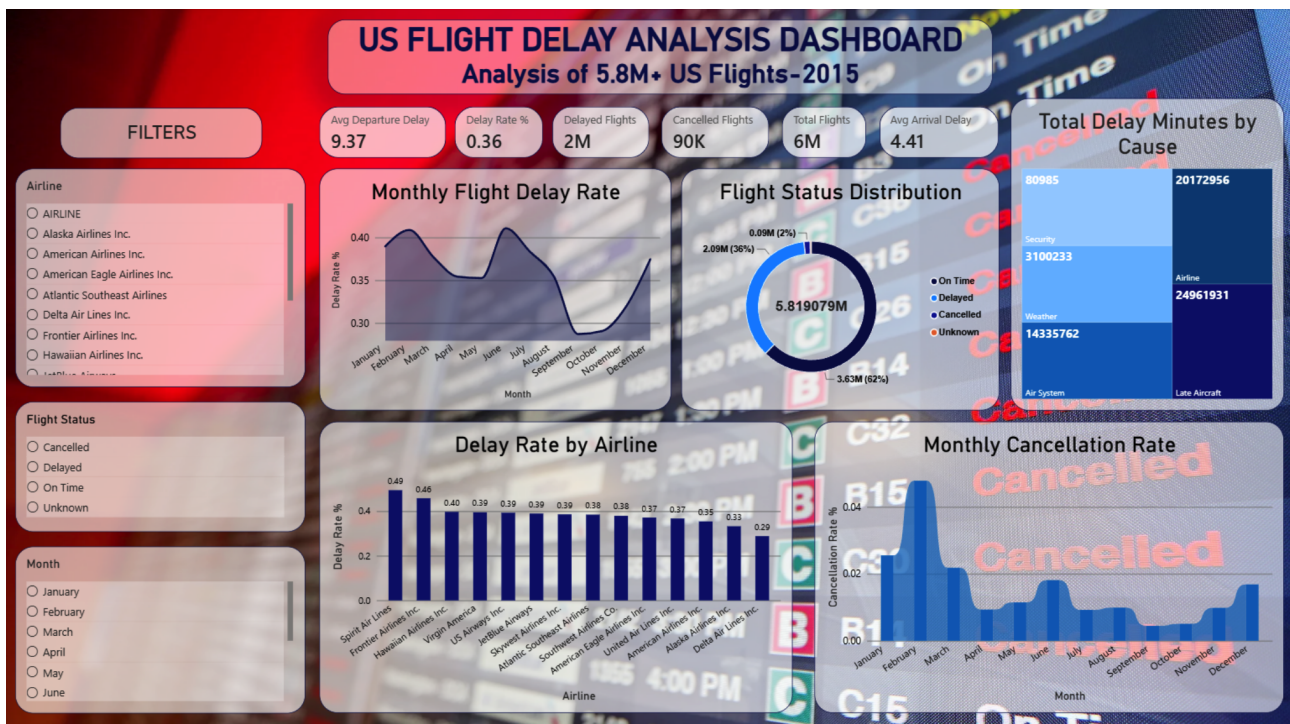


Figure 1: Power BI dashboard overview

Dashboard Screenshot — Airport and Route Analysis

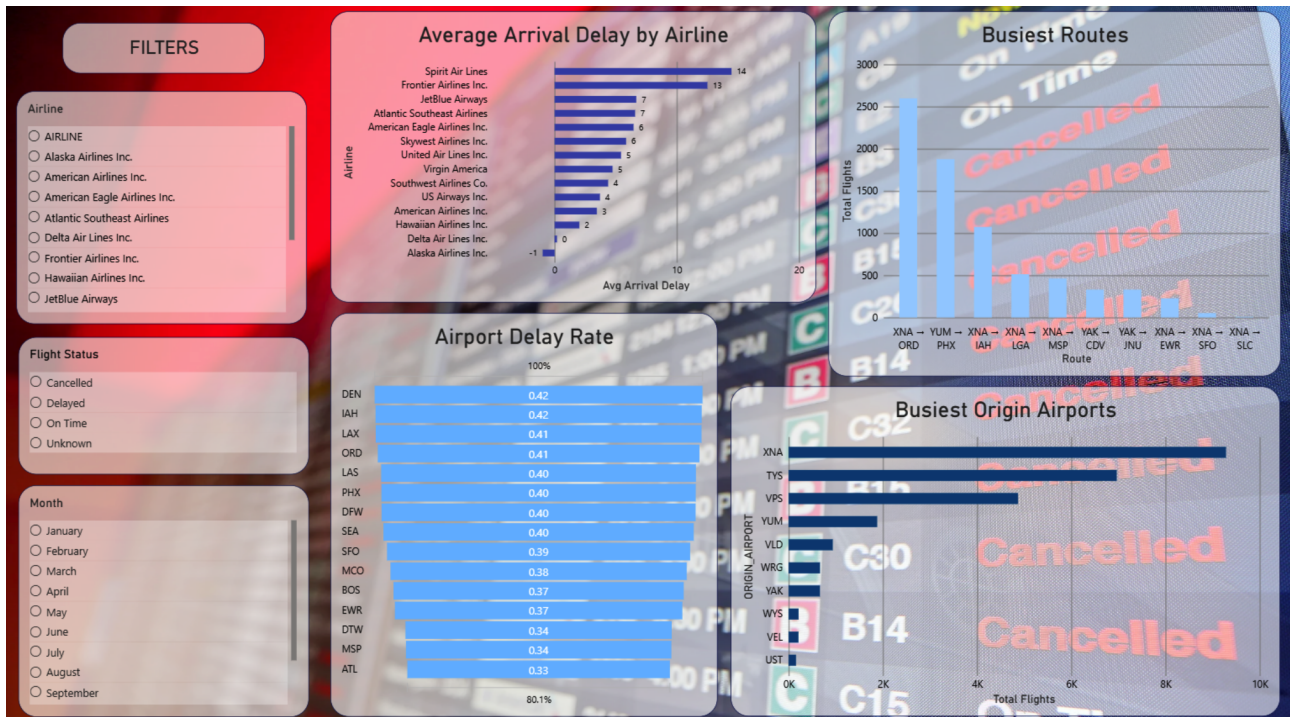


Figure 2: Airport and route analysis views

17. Key Findings

The major findings from the analysis are:

1. The dataset contains **5,819,079 flight records** across 31 columns.
2. Approximately **62.33% of flights were classified as on time**.
3. Approximately **35.86% of flights were delayed**.
4. Approximately **1.54% of flights were cancelled**.
5. The average departure delay was **9.37 minutes**.
6. The average arrival delay was **4.41 minutes**.
7. Delay rates varied considerably across airlines.
8. Late aircraft delays accounted for approximately **24.96 million recorded delay minutes**.
9. Airline-related delays accounted for approximately **20.17 million recorded delay minutes**.
10. Air-system delays accounted for approximately **14.34 million recorded delay minutes**.
11. Monthly delay rates varied considerably, with June at approximately **41.08%** and September at approximately **28.70%**.
12. ATL was the busiest origin airport in the analysis, with **346,836 flights**.
13. The SFO–LAX route was the busiest analyzed route, with **13,400 flights**.
14. The LAS–LAX route had a delay rate of approximately **51.54%** among the listed busiest routes.

18. Business Insights

The analysis provides several useful operational insights.

Airline Operations

Differences in airline delay rates indicate that operational performance varies across carriers. Monitoring delay rates can help identify areas where operational processes may require further investigation.

Aircraft Scheduling

The large number of delay minutes associated with late aircraft suggests that delays can propagate from one flight to subsequent flights.

Air System

Air-system delays affected a large number of flights, demonstrating the importance of air traffic and airport system conditions in overall flight performance.

Seasonal Patterns

The monthly analysis shows that flight performance changes throughout the year. This can be useful for planning resources during periods with higher delay or cancellation rates.

Airport Operations

The airport analysis identifies both high-volume airports and airports with relatively high delay rates. These provide useful areas for deeper operational analysis.

Route Planning

Route-level analysis makes it possible to identify frequently operated routes and examine their corresponding delay rates.

19. Limitations

This project has several limitations.

1. Historical Dataset

The analysis is based on historical 2015 flight data and therefore does not represent current airline performance.

2. Missing Values

Several operational fields contain missing values, particularly delay-cause fields and cancellation reasons.

3. Limited External Factors

The dataset does not provide all possible external factors that could affect flight operations, such as detailed weather conditions, airport staffing levels, aircraft maintenance information, or real-time air traffic conditions.

4. Descriptive Analysis

The project primarily focuses on descriptive and exploratory analytics. It does not attempt to establish causal relationships between individual factors and delays.

5. Large Dataset

The original flights.csv file is very large and therefore was not directly included in the GitHub repository. The project instead provides the dataset source through Kaggle.

20. Future Scope

The project can be extended in several ways.

Predictive Delay Modeling

Machine learning models could be developed to predict whether a future flight will be delayed.

Delay Duration Prediction

A regression model could estimate the expected number of delay minutes.

Real-Time Flight Monitoring

The dashboard could be connected to real-time flight data to monitor current operational performance.

Weather Integration

Weather information could be combined with flight data to investigate the relationship between weather conditions and delays.

Advanced Route Analysis

Future analysis could investigate individual routes by airline, time of day, season, and airport.

Automated Reporting

The Power BI dashboard could be connected to regularly updated data to provide automated operational reports.

21. Conclusion

This project analyzed 5.82 million US flight records to understand flight delays and operational performance.

Using Python, the dataset was explored, cleaned, and analyzed across multiple dimensions including flight status, airlines, delay causes, months, airports, and routes.

The analysis showed that approximately 35.86% of flights were delayed, while the average departure delay was 9.37 minutes and the average arrival delay was 4.41 minutes.

The delay-cause analysis identified late aircraft, airline-related issues, and air-system delays as major contributors to recorded delay minutes.

The project also identified differences in airline delay rates, monthly patterns, busy airports, airport delay rates, and frequently operated flight routes.

Finally, Power BI was used to convert the analytical results into an interactive dashboard, providing a visual way to explore flight operational performance.

Overall, the project demonstrates how data analytics and business intelligence tools can be used to transform a large transportation dataset into meaningful operational insights.

22. References

Dataset

2015 Flight Delays and Cancellations Dataset — Kaggle

Dataset link: [add the Kaggle URL from your GitHub README here]

Tools

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Power BI
- Kaggle
- GitHub

Appendix A — Project Files

The project repository contains the following major files:

```
Airline-Flight-Delay-Operational-Performance-Analytics/  
|-- README.md  
|-- airlines.csv  
|-- flight-delay-analysis.ipynb  
|-- dashboard_overview.png  
`-- dashboard_airport_routes.png
```

The Power BI *.pbix* file is hosted externally because its file size exceeded GitHub's standard upload limit.